Predicting Red Wine Quality Score

A Physicochemical Approach using Machine Learning

GROUP: #15

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PRESENTATION

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– MEET OUR TEAM –

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Exploratory Data Analysis
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Classification Models
(Decision Tree, KNN, Naive Bayes, LR)

CS 418 - Group #15

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01

Introduction

Why Predict Wine Quality?

The Problem

Expert tasting panels are **subjective and costly.**

The global wine market was valued at **$489 billion in 2023**, expected to reach $825B by 2032 (IMARC Group, 2024).

Quality certification bottlenecks affect **millions of bottles annually.**

Portugal alone exported **over 3 million hectoliters** of wine in 2022 (IVV Portugal).

Our Solution

- Use physicochemical lab measurements as predictors
- Train ML models to automate quality scoring
- Task 1 - Regression: predict exact quality score (3-8)
- Task 2 - Classification: predict Good (>=7) vs Not Good (<7)

Dataset Overview

1,599	11	3 - 8	0
Wine Samples	Input Features	Quality Score Range	Missing Values

Physicochemical Features (Predictors)

- Fixed acidity
- Volatile acidity
- Citric acid
- Residual sugar
- Chlorides
- Free sulfur dioxide
- Total sulfur dioxide
- Density
- pH
- Sulphates
- Alcohol

Source

UCI Machine Learning Repository

Vinho Verde Red Wine Dataset

(Cortez et al., 2009)

Target Variable:

quality: integer score assigned by wine experts (scale 0-10)

02

Descriptive Statistics

Summary Statistics

Feature	Mean	Std	Min	Q1	Median	Q3	Max
fixed acidity	8.32	1.74	4.60	7.10	7.90	9.20	15.90
volatile acidity	0.528	0.179	0.120	0.390	0.520	0.640	1.580
citric acid	0.271	0.195	0.000	0.090	0.260	0.420	1.000
residual sugar	2.539	1.410	0.900	1.900	2.200	2.600	15.500
chlorides	0.087	0.047	0.012	0.070	0.079	0.090	0.611
free SO2	15.87	10.46	1.00	7.00	14.00	21.00	72.00
total SO2	46.47	32.90	6.00	22.00	38.00	62.00	289.00
density	0.9967	0.0019	0.990	0.996	0.9968	0.9978	1.004
pH	3.311	0.154	2.740	3.210	3.310	3.400	4.010
sulphates	0.658	0.170	0.330	0.550	0.620	0.730	2.000
alcohol	**10.42**	**1.07**	**8.40**	**9.50**	**10.20**	**11.10**	**14.90**
quality	5.636	0.808	3.00	5.00	6.00	6.00	8.00

Alcohol highlighted - strongest predictor of quality (r = 0.48)

03

Data Visualisation

Quality Score Distribution & Correlation

Quality Score Distribution

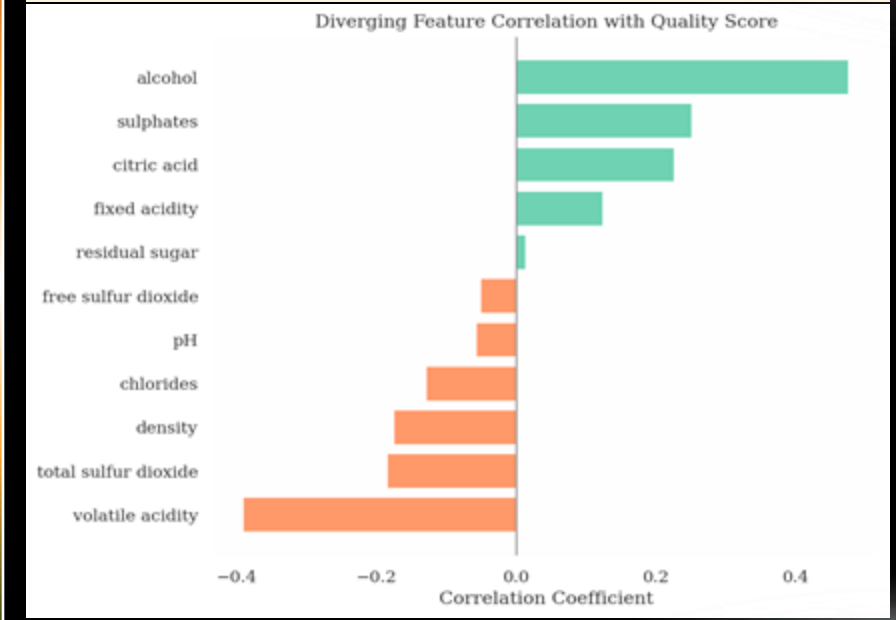
Scores cluster at 5-6; very few wines reach 8 or score below 4.

Correlation Heatmap

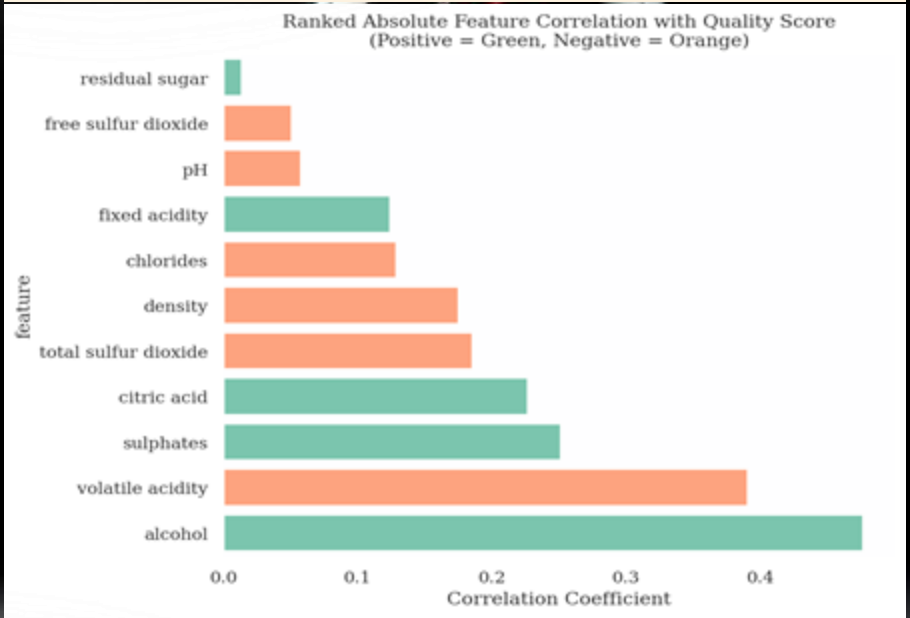
Alcohol (r=0.48) and volatile acidity (r=-0.39) are the strongest correlates of quality.

Feature Importance by Correlation with Quality

Diverging Correlation



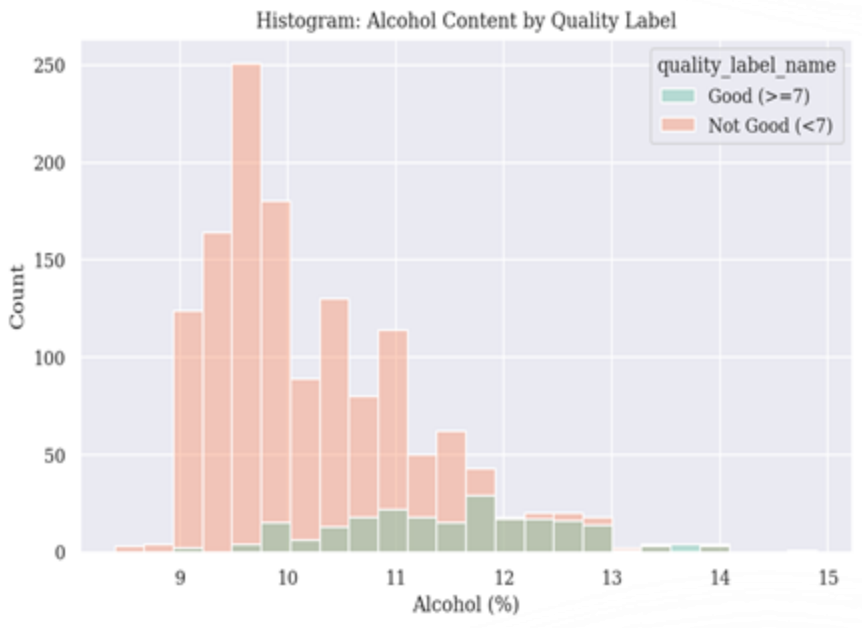
Ranked Absolute Correlation



Key insight: alcohol & volatile acidity dominate. Residual sugar, free SO2, and pH show minimal signal.

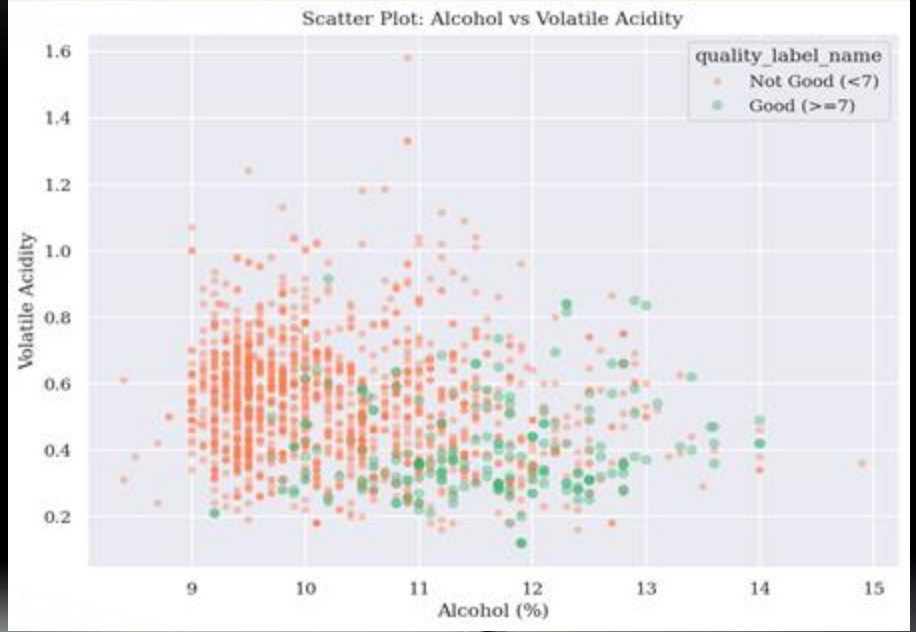
Alcohol Distribution & Scatter Plot

Histogram - Alcohol by Quality Label



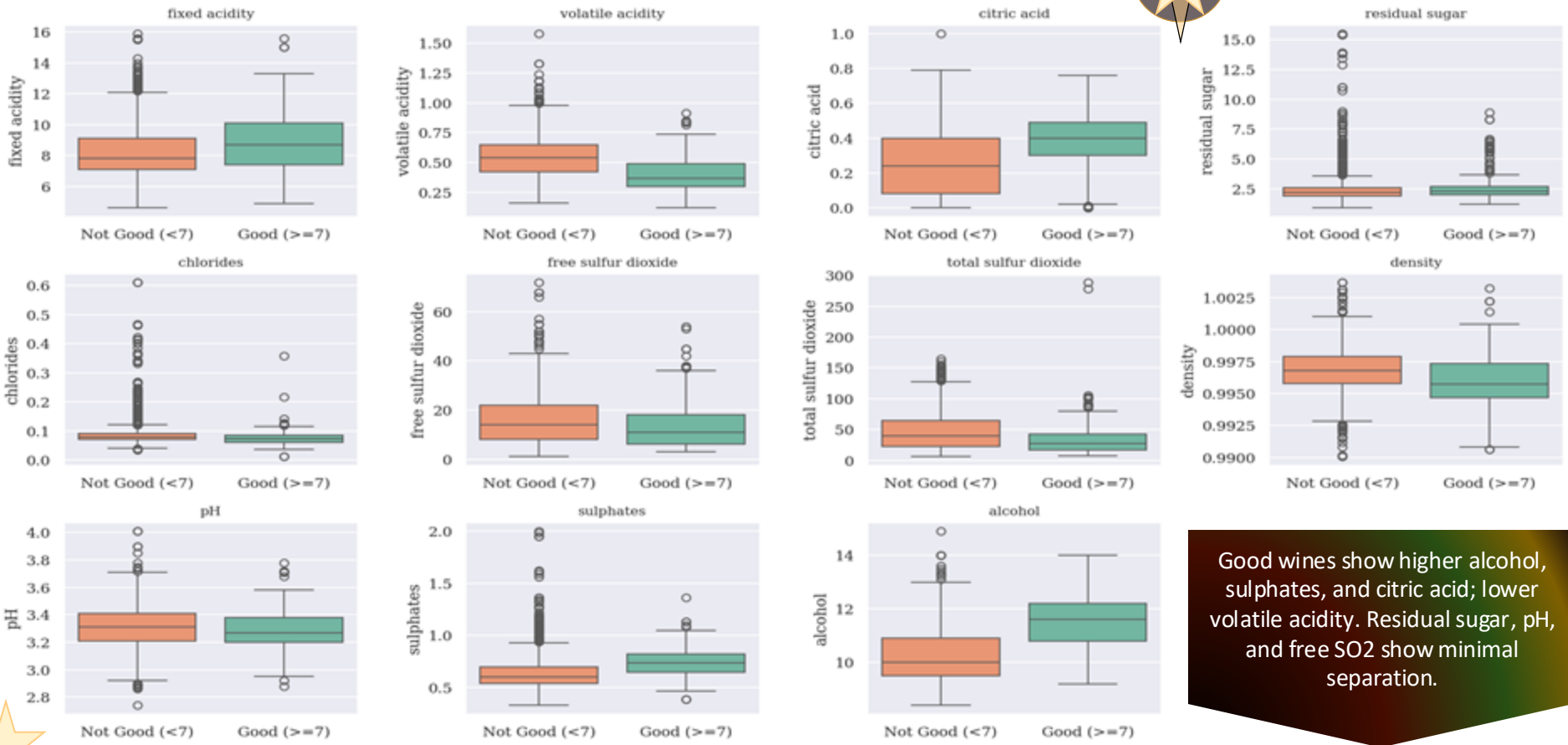
Good wines skew toward higher alcohol content.

Scatter - Alcohol vs Volatile Acidity



Good wines cluster top-right; class overlap explains model difficulty.

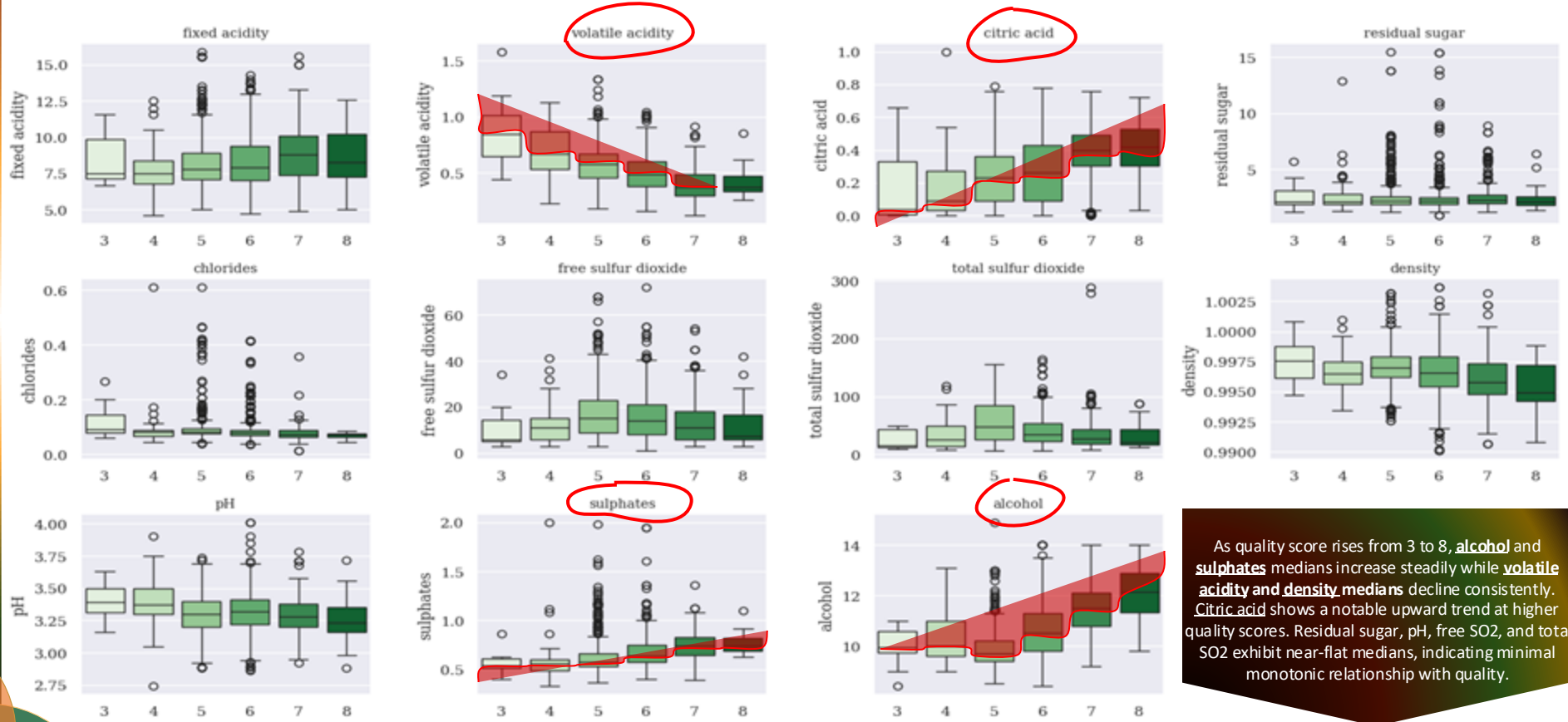
Box Plots - All Features by Quality Label



Good wines show higher alcohol, sulphates, and citric acid; lower volatile acidity. Residual sugar, pH, and free SO2 show minimal separation.

Box Plots - All Features by Quality Score - Trend in Median .

(Trends in Median Values as Q-Score rise)



As quality score rises from 3 to 8, **alcohol** and **sulphates** medians increase steadily while **volatile acidity** and **density medians** decline consistently. Citric acid shows a notable upward trend at higher quality scores. Residual sugar, pH, free SO2, and tota SO2 exhibit near-flat medians, indicating minimal monotonic relationship with quality.

04

Regression

Linear Regression - Predicting Quality Score

Setup

Predicted feature:
quality score (integer 3-8)

5 feature combos tested:

C1: All 11 features

C2: Top 3 predictors

C3: Flavor profile

C4: Top 5 by correlation

C5: Low-collinearity set

Metric: RMSE, R2
80/20 train/test split
StandardScaler applied

Combo	Name	Train RMSE	Test RMSE	Train R2	Test R2
C1	All 11 features	0.6409	0.6687	0.3484	0.3916
C2	Top 3 predictors	0.6557	0.6694	0.3181	0.3903
C3	Flavor profile	0.6610	0.6896	0.3069	0.3530
C4	Top 5 by corr.	0.6497	0.6756	0.3305	0.3790
C5	**Low-collinearity**	**0.6542**	**0.6683**	**0.3211**	**0.3922**

Best model: C5 (Low-collinearity set) - Test R2 = 0.392, Test RMSE = 0.668.
All combos performed similarly - alcohol, volatile acidity, and sulphates carry most predictive signal.

Regression Results - Visualised

Key Takeaways

R2 ~ 0.35-0.39 across ALL combos

Adding features beyond top 3 yields marginal gain only.

Quality scores are integer-valued and clustered - a hard ceiling for linear models.

Non-linear methods (e.g. Random Forest) expected to perform better. But within the scope of CS 418, we chose only methods that were covered in class.

MSE ~ 0.447 | RMSE ~ 0.669 | R2 ~ 0.392 (best model, C5 Low-collinearity set on test set)

05

Classification

Classification - Predicting Good vs Not Good Wine

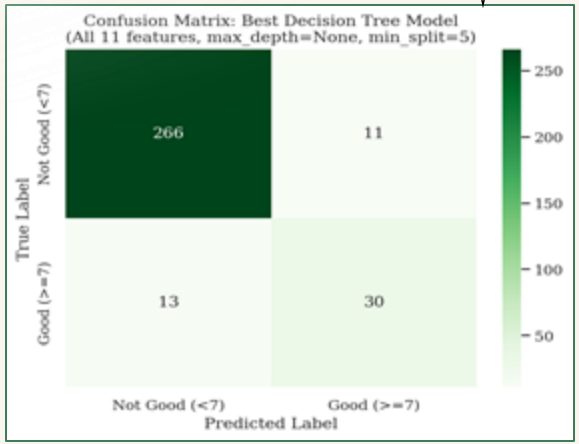
Predicted: quality_label Good (>=7) = 1 Not Good (<7) = 0
Class imbalance: 86.4% Not Good / 13.6% Good

4 classifiers tested - 2 feature sets each: All 11 features and Top 5 by correlation

1	2	3	4
Decision Tree	**KNN**	**Naive Bayes**	**Logistic Regr.**
Tuned: max_depth, min_samples_split	Tuned: k = 3, 5, 7, 11	Gaussian NB - no tuning needed	max_iter=1000, L2 regularisation

Decision Tree - Best Classifier

Max Depth	Min Split	Feature Set	Accuracy	Precision	Recall	F1 Score
None	**5**	**All 11**	**0.9250**	**0.7317**	**0.6977**	**0.7143**
None	2	All 11	0.9250	0.7317	0.6977	0.7143
None	10	All 11	0.9188	0.7180	0.6512	0.6829
10	2	All 11	0.9125	0.6923	0.6279	0.6585
5	5	All 11	0.9063	0.7097	0.5116	0.5946
None	5	Top 5	0.8844	0.5682	0.5814	0.5747



92.5% Accuracy

73.2% Precision

69.8% Recall

71.4% F1 Score

Best config: max_depth=None, min_samples_split=5, All 11 features.
Correctly classified 266/277 not-good and 30/43 good wines.

KNN & Naive Bayes Results

K-Nearest Neighbors

K	Feature Set	Accuracy	Precision	Recall	F1
3	Top 5	0.8719	0.5313	0.3953	0.4533
5	Top 5	0.8656	0.5000	0.3256	0.3944
3	All 11	0.8406	0.3889	0.3256	0.3544
7	All 11	0.8719	0.5500	0.2558	0.3492

Best F1: 0.453 with k=3, Top 5 features. Extra dimensions add noise dimensionality hurts KNN here.

Gaussian Naive Bayes

Feature Set	Accuracy	Precision	Recall	F1
All 11 features	0.8656	0.5000	0.7209	0.5905
Top 5 by corr.	0.8844	0.5682	0.5814	0.5747

Highest recall (0.721) - best at catching good wines despite class imbalance.
Trade-off: lower precision (0.500).

Naive Bayes vs KNN:

NB F1 = 0.59 vs KNN F1 = 0.45

NB is significantly better at identifying minority class (Good wines).

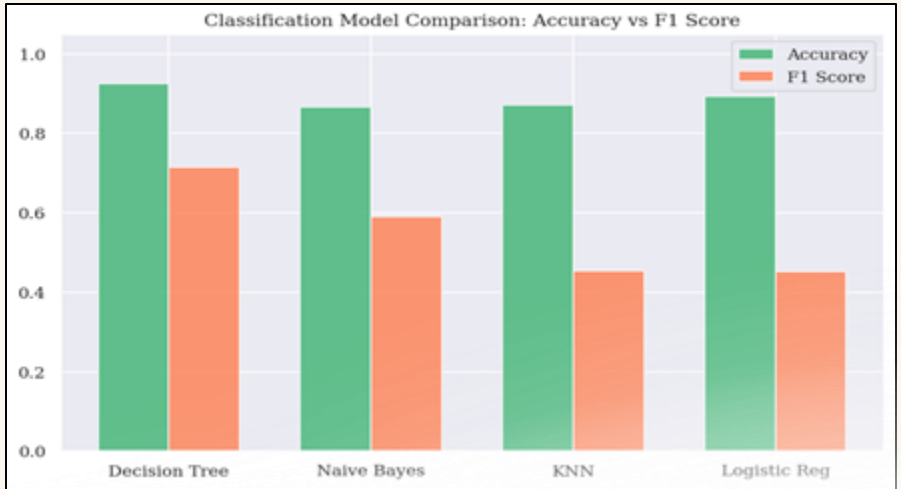
Use NB when recall matters more than precision.

Logistic Regression & Model Comparison

Logistic Regression

Feature Set	Accuracy	Precision	Recall	F1
All 11 features	0.8938	~0.73	~0.33	~0.45
Top 5 by corr.	~0.884	~0.57	~0.46	~0.51

High accuracy, low recall - linear boundary struggles with imbalanced minority class. Similar F1 to KNN.



All Models - Best Results

Model	Accuracy	Precision	Recall	F1
Decision Tree	**0.925**	**0.732**	**0.698**	**0.714**
Naive Bayes	0.866	0.500	0.721	0.590
KNN	0.872	0.531	0.395	0.453
Logistic Reg.	0.894	~0.73	~0.33	~0.45

Winner: Decision Tree

Best balance of precision and recall.
92.5% accuracy - highest F1 (0.714).

If recall is priority:
Naive Bayes (F1=0.59, Recall=0.72).

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Conclusion

What We Learned

Regression

- Linear regression explains ~35-39% of variance (R2 ~ 0.35-0.39)
- RMSE ~ 0.67-0.69 across all feature combos
- Top 3 features (alcohol, volatile acidity, sulphates) carry most linear signal
- Integer-valued, clustered scores create a ceiling for linear models
- Non-linear models (e.g. Random Forest) expected to perform significantly better

Classification

- Decision Tree wins: 92.5% accuracy, F1 = 0.714
- Naive Bayes best for recall (0.721) - important if missing good wines is costly
- KNN and Logistic Regression: F1 ~ 0.45, weaker minority class detection
- 86%/14% class imbalance makes F1 a better metric than accuracy alone

Key Insight

- Alcohol is the #1 predictor of wine quality (r = 0.48)
- Volatile acidity is the #1 negative predictor (r = -0.39)
- Classification is a better framing than regression for this dataset
- Future: SMOTE for imbalance, ensemble models, combine red + white datasets

Regression & Classification | CS 418: Introduction to Data Science | Spring 2026

Predicting Red Wine Quality

A Physicochemical Approach using Machine Learning

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Thank You! Q&A?